Low-Resource Moroccan Dialect (Darija) ASR Under Compute Constraints

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Abstract

This report presents a practical low-resource ASR pipeline for Moroccan Darija under strict compute constraints. Two public datasets were selected, enriched with diarization-derived metadata, filtered with a custom usability score, merged, split into train/validation/test, and used to fine-tune two Whisper checkpoints: Whisper Base and Whisper Small. The final system was evaluated with WER and CER, analyzed by multiple metadata buckets, and compared against a public Darija Whisper checkpoint from Hugging Face. The main result is that the locally fine-tuned Whisper Small checkpoint clearly outperformed both the local Whisper Base checkpoint and the evaluated public small checkpoint on the curated test set. The deeper conclusion is that Moroccan Darija ASR is bottlenecked not only by model size, but also by overlap speech, segmentation quality, orthographic instability, code-switching, and dataset contamination.

All my repositories and HF link resulted from this project:

- `Add useful metadata to dataset repo link.`
- `Model training and evaluation repo link.`
- `Dataset 1 HF link.`
- `Dataset 2 HF link.`
- `checkpoint small HF link.`
- `checkpoint base HF link.`

1 Objective

The goal of this work was to fine-tune and evaluate a lightweight automatic speech recognition system for Moroccan Darija under strict compute constraints:

- a single GPU with a budget below 16 GB VRAM (in practice, only 8 GB VRAM were available),
- use of an existing pretrained ASR model,
- use of a relatively small labeled dataset,
- limited implementation time.

The main question was not only which model performs better, but also what the real bottlenecks are when building ASR for Moroccan Darija: limited curated data, overlap speech, orthographic instability, dialect variation, and code-switching.

2 Why Moroccan Darija and Why Whisper

2.1 Why Moroccan Darija

Moroccan Darija is an important but difficult low-resource ASR setting because it suffers from several structural issues:

- relatively limited high-quality labeled speech data,
- scarce linguistic resources,

- strong dialectal variation,
- lack of a single standardized writing system,
- frequent code-switching with French, Spanish, and sometimes English,
- a speech-first culture, with less stable written conventions.

These properties make Darija a strong test case for data curation decisions, not only model choice. In other words, this assignment is as much about deciding *what to train on* as it is about deciding *what to train*.

2.2 Why Whisper

Given the timeline and the hardware limits, Whisper was the most practical choice.

Compared with alternatives such as Wav2Vec2-CTC or Conformer-based models, Whisper offered the best trade-off for this assignment because:

- it has a strong multilingual prior,
- it adapts well in low-resource settings,
- it is easy to fine-tune and evaluate quickly with the Hugging Face ecosystem,
- base and small checkpoints are realistic under limited GPU memory,
- it supports seq2seq transcription without extra engineering for a CTC vocabulary or external decoder pipeline.

With only 8 GB VRAM available, larger models were not realistic. The practical candidates were therefore lighter Whisper checkpoints, and I focused on:

- Whisper Base,
- Whisper Small.

This also enabled a clean comparison between two realistic model sizes under the same resource budget.

3 Dataset Selection

3.1 Selected Datasets

I selected the following two datasets because together they provide more diversity than a single-source corpus, especially in terms of speaker variety and conversational structure:

- `adiren7/darija_speech_to_text`
- `KandirResearch/DarijaTTS-clean`

After processing, I published the metadata-enriched versions here:

- `EtMmohammedHafsati/darija_tts_clean_metadata_full`
- `EtMmohammedHafsati/darija_speech_to_text_metadata_full`

3.2 Why These Two

The choice was motivated by:

- diversity of male and female speakers,
- more than one speaker in many recordings,
- complementary profiles:
  - many shorter samples from the TTS-clean source,
  - many longer samples from the speech-to-text source.

This diversity is useful for ASR, but it also introduces a problem: many samples contain overlap speech or multi-speaker behavior.

3.3 Rejected Datasets

Several other Moroccan Darija datasets were considered but not used:

- team1group/hdrtna_darija: too much code-switching,
- atlasia/Moroccan-Darija-Wiki-Audio-Dataset: same-speaker bias and heavy code-switching,
- one additional dataset was discarded because of frequent audio/transcript mismatches and low quality,
- anaszil/Segmented-Moroccan-Darija-Wiki-Audio-Dataset: cleaner, but mostly same-speaker, which risks speaker bias.

This filtering step was important. For this assignment, the problem was not only to find data, but to find data that is trainable under limited time and limited compute.

4 Data Curation with Diarization and Usability Scoring

4.1 Motivation

The two selected datasets were not clean enough to be used blindly. A large number of clips contained:

- overlap speech,
- more than one active speaker,
- rapid turn-taking,
- segments that are not ideal for standard single-speaker ASR.

Instead of discarding all conversational audio, I used a diarization-based metadata pipeline to characterize each sample and decide whether it was suitable for training.

Two hypotheses guided the curation:

1. **Backchannel responses can still be useful training examples.** A clip does not need to be perfectly single-speaker to be useful. Short acknowledgments or mild interaction may still preserve a dominant transcribed speaker.
2. **A dominant speaker can make a multi-speaker clip usable.** If one speaker clearly dominates the speech time, the clip may still behave like a usable ASR example.

So the objective was not to enforce an unrealistically pure dataset, but to keep clips that remain ASR-tractable.

5 Metadata Added Per Sample

For each sample, I generated the following metadata fields:

- `duration_sec`
- `num_speakers`
- `multiple_speakers`
- `speaker_turns`
- `turns_per_minute`
- `dominant_speaker_ratio`
- `second_speaker_ratio`
- `non_dominant_speech_ratio`

- `speaker_balance_score`
- `speaker_entropy`
- `overlap_speech`
- `overlap_duration_sec`
- `overlap_ratio`
- `num_overlap_regions`
- `mean_overlap_region_sec`
- `max_overlap_region_sec`
- `max_concurrent_speakers`
- `asr_usability_score`
- `asr_usable_single_speaker`
- `source_split_original`
- `metadata_generated_by`
- `processing_error`

These features were used both for filtering and for later analysis.

6 Explanation of the Custom Usability Score

6.1 Speaker Statistics

From diarization segments, I computed speaker-structure features such as:

- number of speakers,
- number of speaker turns,
- dominant speaker ratio,
- second speaker ratio,
- non-dominant speech ratio,
- speaker entropy,
- speaker balance score.

The most important intuition is:

- if one speaker occupies most of the speech time, the clip is easier for standard ASR,
- if speech is evenly distributed across speakers, the clip is less ASR-friendly,
- if overlap is high, it becomes harder again.

6.2 Score Definition

The custom usability score is defined in the interval $[0, 1]$, where higher means more suitable for standard single-speaker ASR.

First, turn-taking density is defined as:

$$\text{turns_per_minute} = \frac{\text{speaker_turns}}{\text{duration_sec}/60}$$

Then a bounded penalty is applied:

$$\text{turns_penalty} = \min\left(\frac{\text{turns_per_minute}}{40}, 1\right)$$

Finally, the score is:

$$\text{score} = 0.6 \cdot (1 - \min(\text{overlap_ratio}, 1)) + 0.3 \cdot \text{dominant_speaker_ratio} + 0.1 \cdot (1 - \text{turns_penalty})$$

and then clipped to $[0, 1]$.

6.3 Interpretation of the Score

This weighting reflects the design assumptions of the task:

- **60% overlap term**: overlap is the strongest penalty because simultaneous speech directly harms transcription quality.
- **30% dominant-speaker term**: a clip is rewarded when one speaker clearly dominates, even if another speaker is present briefly.
- **10% turn-rate term**: rapid turn-taking is penalized, but less strongly than overlap.

So a clip receives a high score when:

- overlap is low,
- one speaker dominates most of the speech time,
- turn-taking is not too dense.

Conversely, a clip receives a lower score when:

- overlap is high,
- speech is shared more equally across several speakers,
- speaker changes are too frequent.

6.4 Threshold Used

I kept only samples with:

$$\text{usability score} > 0.85$$

Throughout the report, I refer to this as a *0.85 threshold* or *85% usability threshold*. This was a strict decision that prioritized trainability over maximum data volume.

7 Data Preparation Pipeline

The two selected datasets were merged, deduplicated, filtered, split, normalized, and exported to 16 kHz audio paths for Whisper.

7.1 Split Creation

The merged dataset was organized into:

- 80% train,
- 10% validation,
- 10% test.

7.2 Preparation Statistics

From the preprocessing logs:

Table 1: Raw merged dataset by split

Split	Samples	Hours
Train	18,870	47.99
Validation	2,359	5.94
Test	2,360	5.99

After deduplication:

- total rows: 23,589 $\rightarrow$ 23,551

Rejected samples:

- `text_empty`: 0
- `remove_short_audio_long_text`: 0
- `remove_token_outlier`: 0
- `usability_too_low` ($\leq$ `0.85`): 4,854

Final trainable summary:

Table 2: Final usable data after filtering

Split	Total	Trainable / Whisper-ready
Train	18,833	14,965
Validation	2,359	1,854
Test	2,359	1,878

7.3 Hours Retained

The usable data retained after the 0.85 threshold were:

Table 3: Hours before and after filtering

Split	Raw merged hours	Whisper-ready hours	Median duration (sec)
Train	47.99	34.08	3.39
Validation	5.94	4.00	3.27
Test	5.99	4.17	3.39

Overall:

- total raw audio: about 59.92 hours,

- total whisper-ready audio: about 42.25 hours.

So the strict usability threshold removed:

- about 20.6% of samples,
- about 29.5% of total audio hours.

7.4 Audio Export Validation

All retained examples were exported successfully to 16 kHz WAV:

- train: 14,965 / 14,965 success,
- validation: 1,854 / 1,854 success,
- test: 1,878 / 1,878 success.

The loaded duration from the waveforms matched metadata almost perfectly, with a loaded-vs-metadata hour ratio of approximately 1.000004. This validates the export pipeline.

8 Exploratory Data Analysis

8.1 Sample Count by Source

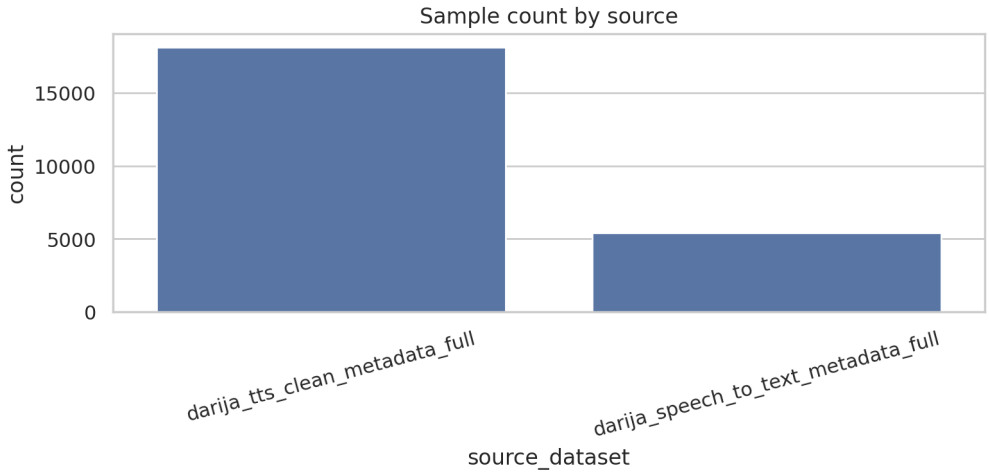

Figure 1: Sample count by source. The TTS-clean source dominates by sample count, while the speech-to-text source contributes fewer but generally longer examples.

The merged dataset is imbalanced by *sample count*: `darija_tts_clean_metadata_full` contributes many more utterances than `darija_speech_to_text_metadata_full`. This matters during training because the model sees many more short TTS-style examples.

8.2 Duration Distribution by Source

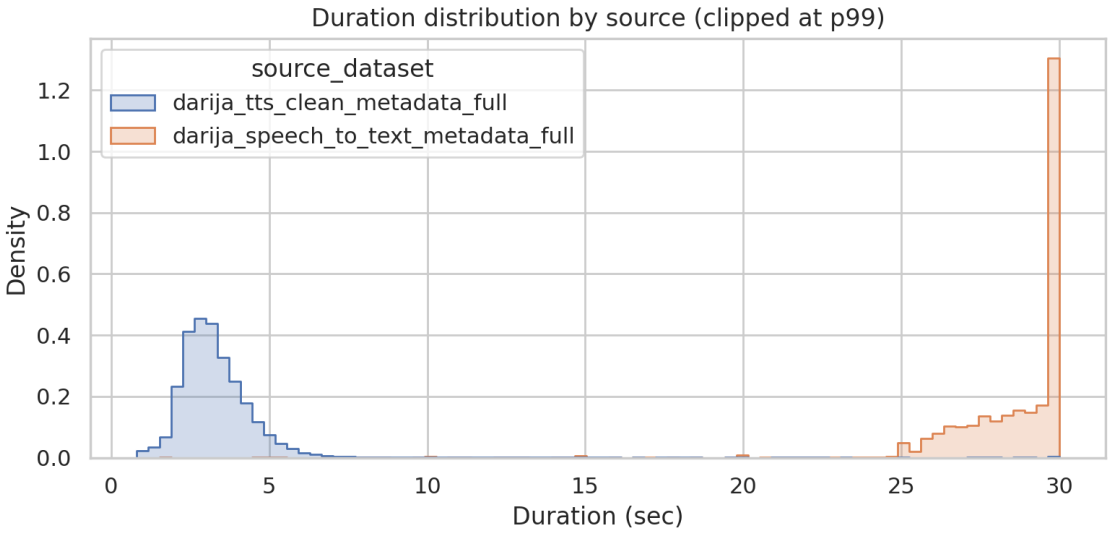

Figure 2: Duration distribution by source, clipped at the 99th percentile. The distribution is clearly bimodal: short clips from the TTS-clean source and long clips from the speech-to-text source.

This is one of the most important plots in the whole analysis. The duration distribution is clearly bimodal:

- the TTS-clean source mostly concentrates around 2–5 seconds, with a peak near 3–4 seconds,
- the speech-to-text source heavily concentrates around 25–30 seconds.

There are very few examples between 6 and 25 seconds. This is concerning for interpretation because the training data are not distributed smoothly over duration. The model is exposed mostly to two regimes:

- short utterances,
- long chunk-like utterances.

This is important when interpreting the duration-based evaluation buckets.

8.3 Custom Usability Score Distribution

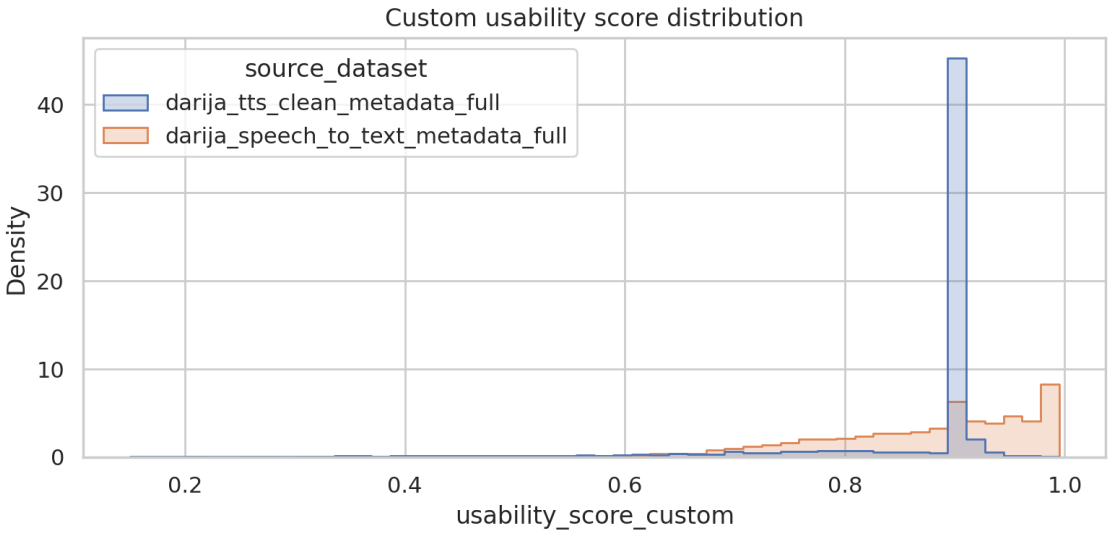

Figure 3: Custom usability score distribution by source. The TTS-clean source is tightly concentrated near the high-score region, while the speech-to-text source is more spread out.

The custom usability score distribution shows:

- the TTS-clean source is tightly concentrated near the upper end,
- the speech-to-text source is more spread out, indicating more conversational and structural variability.

This supports the intuition that the TTS-clean data are more structurally consistent, while the speech-to-text source is harder but more diverse.

8.4 Existing ASR Usability vs Custom Usability

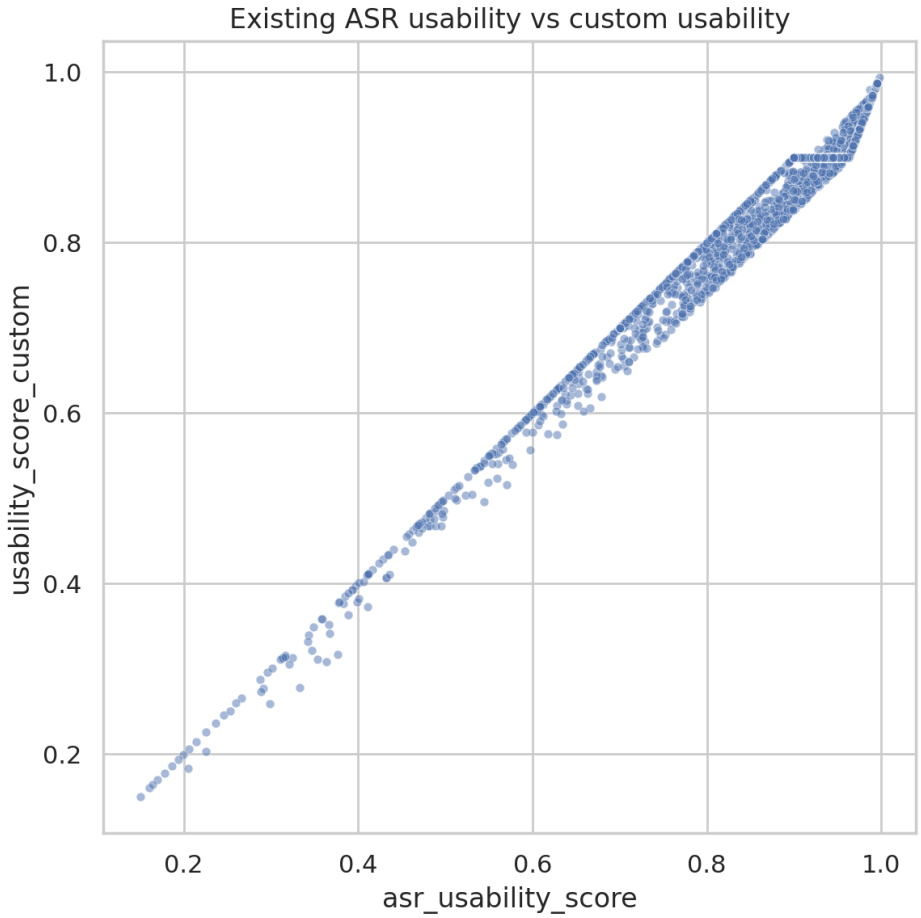

Figure 4: Existing ASR usability score versus the custom usability score. The two measures are strongly correlated.

The strong monotonic relationship indicates that the custom score is not arbitrary. It is aligned with the existing usability signal, but it is more interpretable because it explicitly encodes overlap, dominant-speaker structure, and turn-taking density.

8.5 Overlap Ratio vs Dominant Speaker Ratio

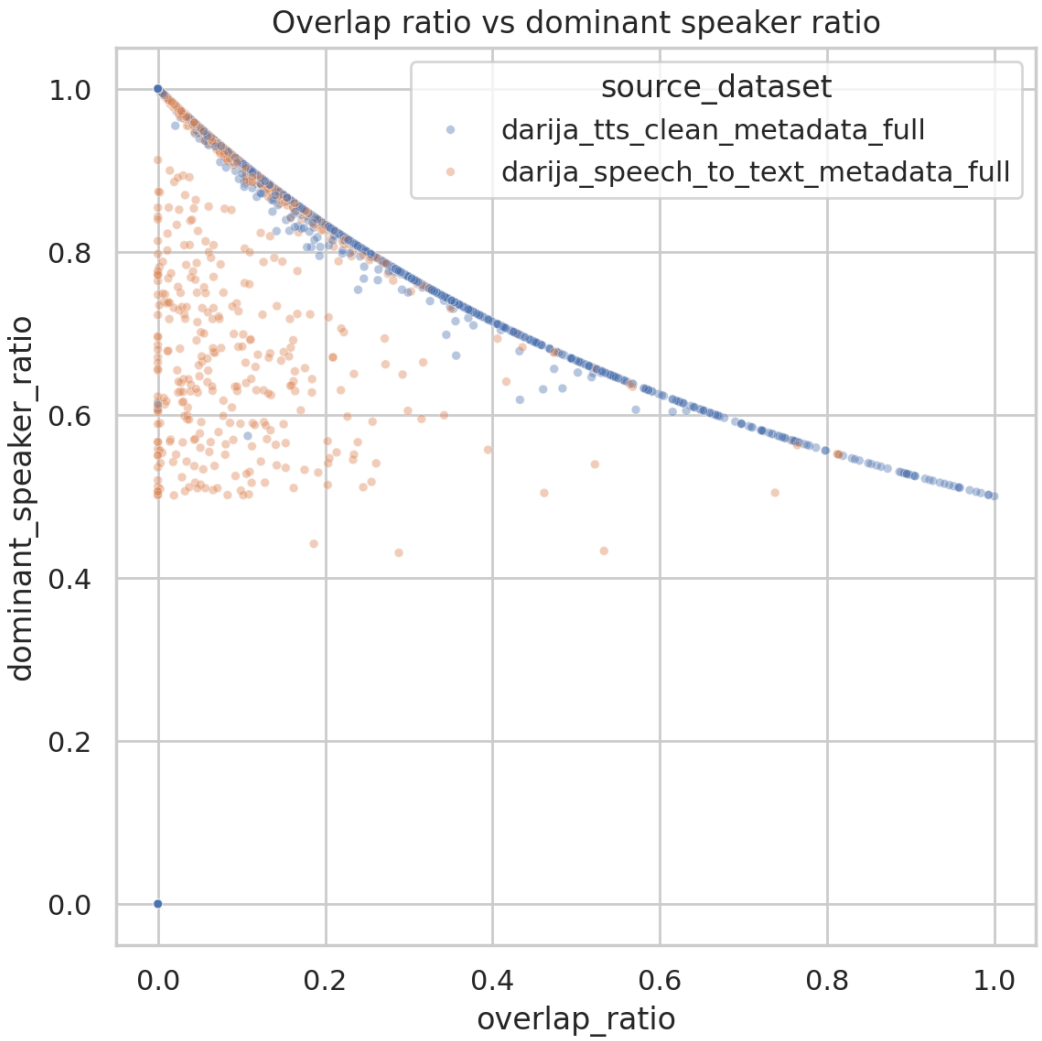

Figure 5: Overlap ratio versus dominant speaker ratio. As overlap increases, dominant-speaker ratio generally decreases.

This plot validates the score design. As overlap increases, the dominant-speaker ratio usually decreases. In other words:

- low-overlap clips often have one clear main speaker,
- higher-overlap clips tend to distribute speech more across multiple speakers.

This is especially clear for the TTS-clean source. The speech-to-text source shows a wider cloud, which is consistent with more conversational variability.

8.6 Quality Bins by Source

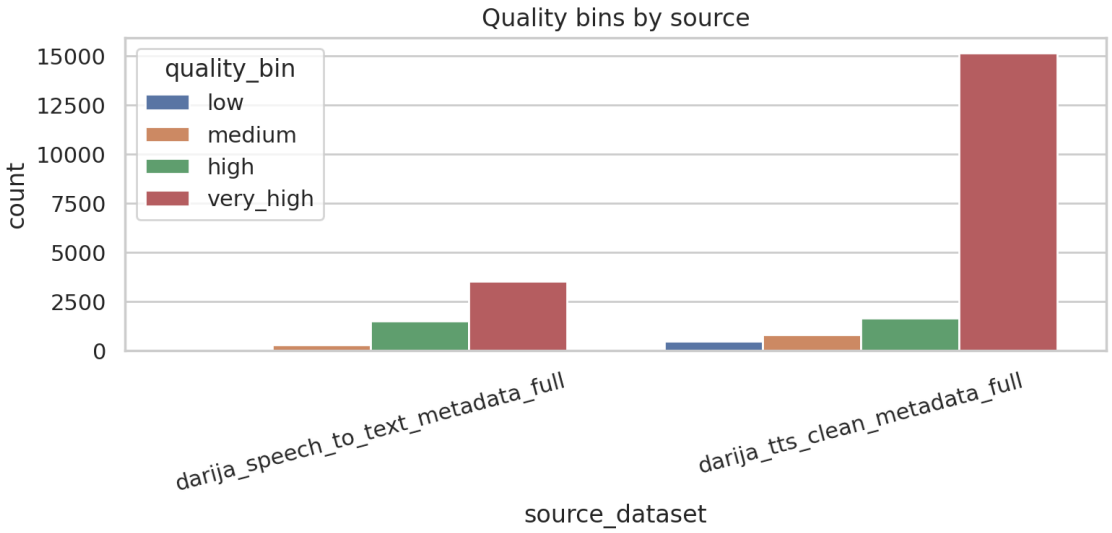

Figure 6: Quality bins by source. Most samples fall in the high or very-high quality bins, especially after filtering.

Most retained samples fall into the upper quality bins. This shows that the filtering process was effective: the final training/evaluation data are not a random subset, but a deliberately curated high-usability subset.

8.7 Tokens per Second by Source

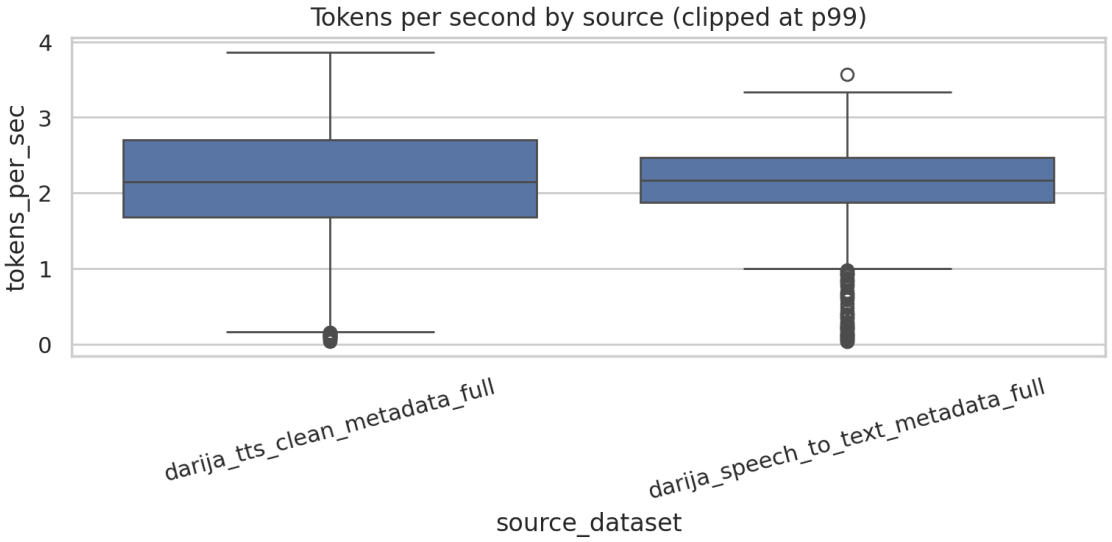

Figure 7: Tokens per second by source, clipped at the 99th percentile. The central speech-rate tendencies are similar, but the sources differ in spread and outliers.

The central speaking-rate tendencies are relatively similar across sources, but the spread and outliers differ. This means difficulty is not only about clip duration; rate-related variability also remains in the corpus.

8.8 Correlation Heatmap

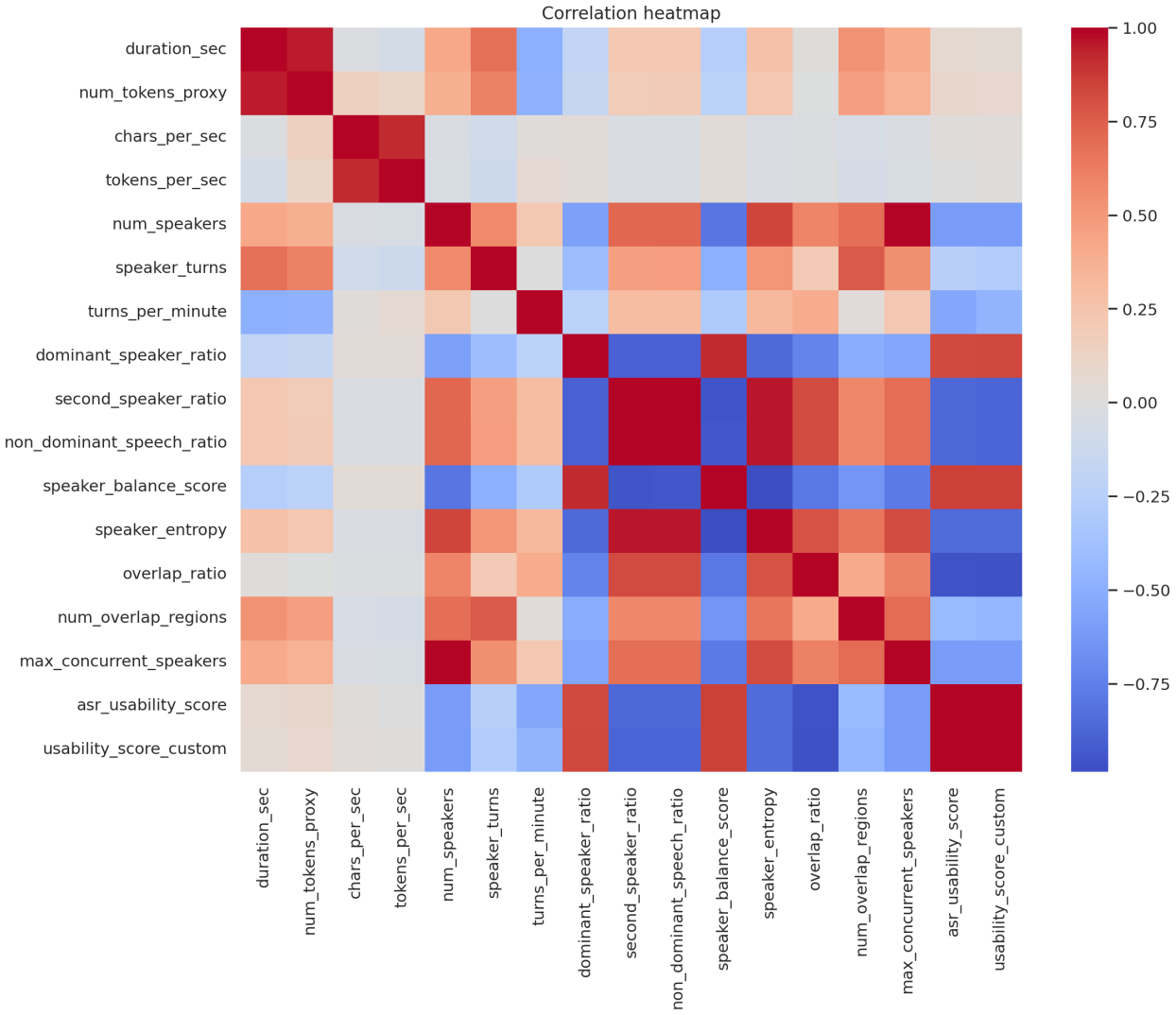

Figure 8: Correlation heatmap for the main metadata features.

The heatmap confirms several intuitive relationships:

- `duration_sec` strongly correlates with `num_tokens_proxy`,
- `chars_per_sec` and `tokens_per_sec` are strongly correlated,
- `asr_usability_score` and `usability_score_custom` are strongly correlated,
- `overlap_ratio` is negatively correlated with dominant-speaker ratio and usability,
- speaker-complexity variables are positively related to overlap and conversational difficulty.

This shows that the metadata are coherent and capture real structure in the corpus.

8.9 Hours and Counts by Processing State

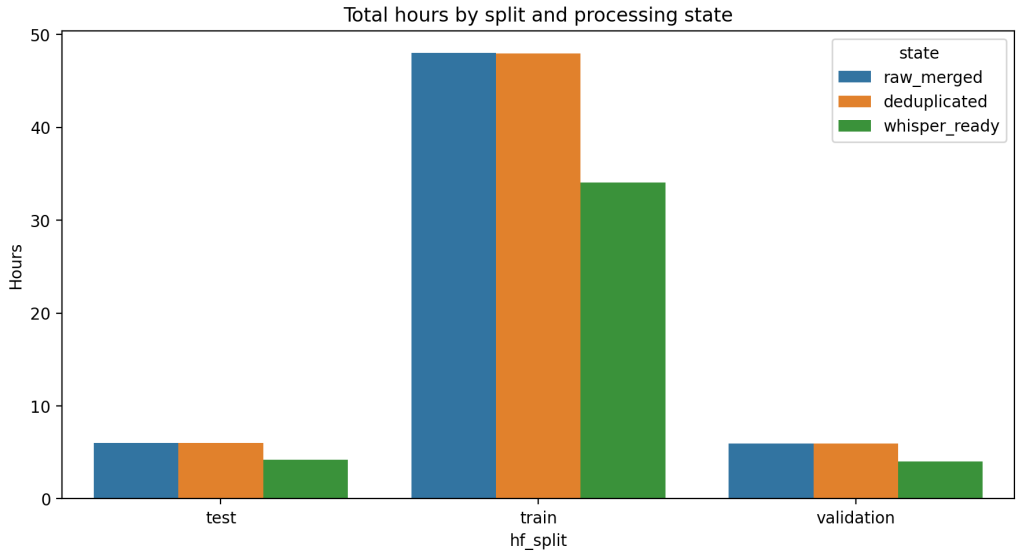

Figure 9: Total hours by split and processing state.

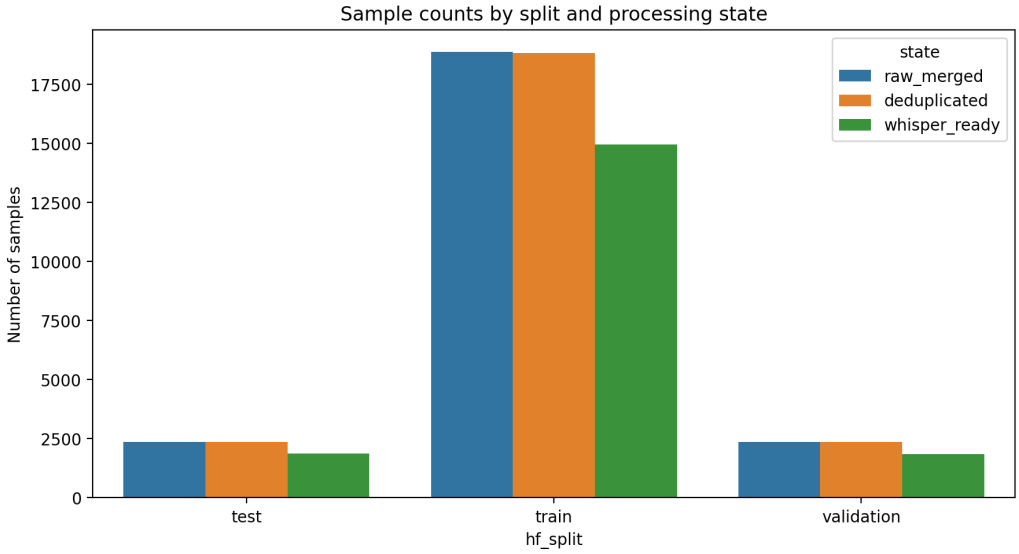

Figure 10: Sample counts by split and processing state.

These plots show that the filtering stage removed a substantial part of the data, but preserved the split structure. The 80/10/10 organization remains consistent after curation.

8.10 Hours and Counts by Split and Source

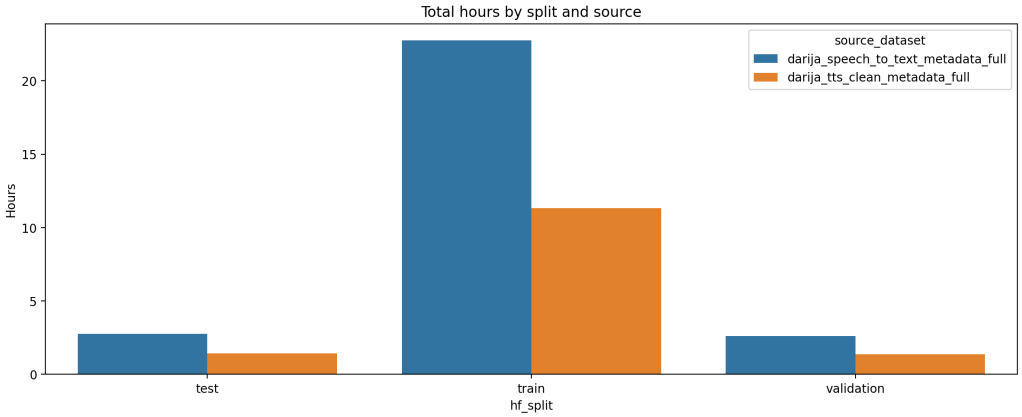

Figure 11: Total hours by split and source. The speech-to-text source contributes fewer samples but more hours.

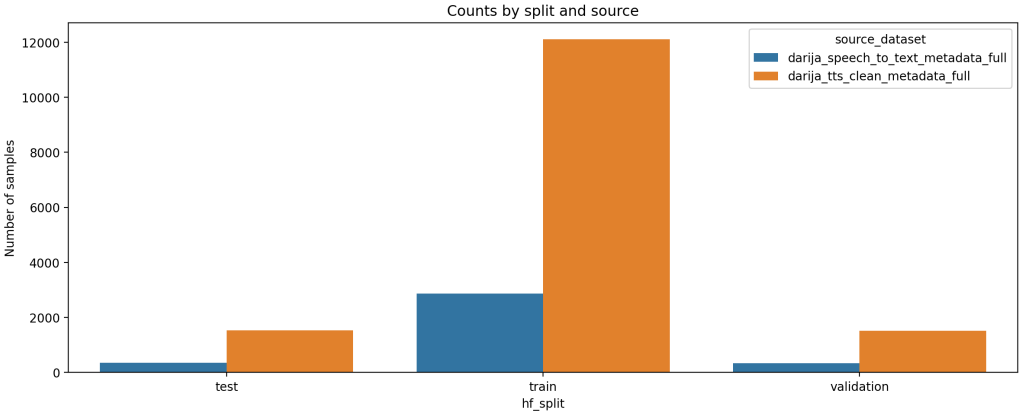

Figure 12: Counts by split and source. The TTS-clean source dominates by count.

These two figures highlight a key asymmetry:

- the TTS-clean source dominates by *count*,
- the speech-to-text source contributes a larger share of *hours*.

This means that the merged corpus is imbalanced in two different ways at once.

9 Transcript Text Normalization

Because Moroccan Darija does not have a fully standardized writing system, transcription quality is affected not only by acoustic difficulty but also by orthographic variation. The same spoken utterance may appear in several written forms depending on the annotator, especially when Arabic script, French influence, punctuation, and informal spelling conventions are mixed. For this reason, I applied a text-normalization step before training and evaluation.

The goal of this normalization was not to aggressively rewrite the language, but to reduce artificial variability that would otherwise inflate WER and CER without reflecting a true ASR failure.

9.1 Motivation

This normalization step was important for three reasons:

- **Orthographic instability:** Moroccan Darija is often written informally, with no single agreed spelling system.
- **Arabic-script variation:** the same word may be written using different Unicode forms or Arabic letter variants.
- **Fairer evaluation:** some prediction/reference mismatches are due more to formatting or spelling conventions than to incorrect recognition.

9.2 Normalization operations

The normalization pipeline included the following operations.

1. Unicode cleanup Control characters and invisible formatting characters were removed. Then the text was normalized with Unicode NFKC normalization in order to collapse compatibility forms into a canonical representation.

2. Whitespace normalization Leading and trailing spaces were removed, and repeated spaces were collapsed into a single space.

3. Arabic character normalization Several Arabic characters were normalized into simpler canonical forms:

- Alef variants were mapped to bare alef: ٱ آ، إ، أ، to ا.
- Hamza-seat forms were normalized: ؤ to و, and ئ to ي.
- Alef maqsura was normalized: ى to ي.
- Some Persian / Urdu variants were mapped to Arabic forms: ک to ك, ے ى، to ي, and ة ه، to ه.
- Lam–alef ligatures were decomposed: ﷲ ﷺ، ﷻ، ﷼، ﷽، ﷹ، ﷸ، ﷷ، ﷶ، ﷵ، ﷴ، ﷳ، ﷲ، ﷱ، ﷰ، ﷯، ﷮، ﷭، ﷬، ﷫، ﷪، ﷩، ﷨، ﷧، ﷦، ﷥، ﷤، ﷣، ﷢، ﷡، ﷠، ﷟، ﷞، ﷝، ﷜، ﷛، ﷚، ﷙، ﷘، ﷗، ﷖، ﷕، ﷔، ﷓، ﷒، ﷑، ﷐، ﷏، ﷎، ﷍، ﷌، ﷋، ﷊، ﷉، ﷈، ﷇ، ﷆ، ﷅ، ﷄ، ﷃ، ﷂ، ﷁ، ﷀ، ﷿، ﷾، ﷽، ﷼، ﷻ، ﷺ، ﷹ، ﷸ، ﷷ، ﷶ، ﷵ، ﷴ، ﷳ، ﷲ، ﷱ، ﷰ، ﷯، ﷮، ﷭، ﷬، ﷫، ﷪، ﷩، ﷨، ﷧، ﷦، ﷥، ﷤، ﷣، ﷢، ﷡، ﷠، ﷟، ﷞، ﷝، ﷜، ﷛، ﷚، ﷙

Therefore, text normalization helps reduce noise in both training and evaluation. It does not solve the deeper linguistic problem of Darija orthographic instability, but it makes the benchmark more consistent and the reported metrics more meaningful.

9.4 Limitations of normalization

This normalization should still be interpreted carefully. It improves consistency, but it does not fully solve:

- true code-switching ambiguity,
- non-standard or phonetic spellings,
- transcript/audio mismatch,
- the lack of a community-wide written standard for Darija.

So normalization reduces evaluation noise, but some residual WER inflation is still expected in Moroccan Darija ASR.

10 Fine-Tuning Setup

I trained two Whisper checkpoints:

- **Whisper Base**, initialized from `openai/whisper-base`
- **Whisper Small**, initialized from `openai/whisper-small`

Checkpoints are available here:

- `EtMmohammedHafsati/whisper-base-darija-merged-diarized`
- `EtMmohammedHafsati/whisper-small-darija-merged-diarized`

This was **full fine-tuning**, not LoRA or adapter tuning. In other words, all model parameters were updated during training. I did not use PEFT, LoRA modules, frozen backbones, or trainable low-rank adapters. The models were trained directly with Hugging Face's `Seq2SeqTrainer`.

10.1 Input pipeline

The training dataset was the curated 16 kHz version saved at:

- `./darija_merged_analysis/whisper_ready_dataset_16k_paths_success_only`

For each sample:

- audio was loaded from `audio_path_16k`,
- resampled to 16 kHz mono with `librosa`,
- converted into Whisper log-Mel input features with `WhisperFeatureExtractor`,
- transcript text was normalized by trimming whitespace and collapsing repeated spaces,
- normalized text was tokenized with the Whisper tokenizer configured for `language="arabic"` and `task="transcribe"`.

The maximum label length was fixed to 225 tokens.

10.2 Model configuration

The Whisper processor and tokenizer were configured for Arabic transcription:

- `language = "arabic"`
- `task = "transcribe"`

During training, the model configuration was set as follows:

- `forced_decoder_ids = None`
- `suppress_tokens = []`
- `use_cache = False`

These settings were used to keep generation behavior explicit and compatible with training under gradient checkpointing.

10.3 Optimization and memory constraints

Training was performed on a single GPU with only 8 GB VRAM available. Because of this hardware limit, I used several memory-saving strategies:

- mixed precision training (`fp16`) when CUDA was available,
- gradient checkpointing,
- gradient accumulation.

The main hyperparameters were:

- per-device train batch size: 8
- per-device evaluation batch size: 8
- gradient accumulation steps: 2
- effective batch size: 16
- learning rate: 1×10^{-5}
- warmup steps: 500
- total training steps: 10,000
- evaluation every 500 steps
- checkpoint save every 500 steps
- logging every 50 steps

The best checkpoint was selected using validation WER, with:

- `metric_for_best_model = "wer"`
- `greater_is_better = False`
- `load_best_model_at_end = True`

10.4 Training duration in epochs

The final curated training split contains 14,965 samples. With an effective batch size of 16, one epoch corresponds to approximately:

$$\frac{14965}{16} \approx 935$$

update steps.

Since each model was trained for 10,000 steps, this corresponds to approximately:

$$\frac{10000}{935} \approx 10.7$$

epochs.

So the experiment note "5000 $\times$ 2 steps" can be interpreted as roughly **10.7 epochs** for each model on the final curated dataset.

10.5 Training objective

The models were fine-tuned with supervised sequence-to-sequence learning using paired audio and transcript data. Evaluation during training used generation-based decoding so that WER and CER reflected the real transcription behavior of the model rather than token-level teacher-forced loss alone.

Overall, this setup should be described as **full-parameter Whisper fine-tuning under memory constraints**, not LoRA or parameter-efficient fine-tuning.

11 Evaluation Protocol

All models were evaluated on the same held-out curated merged test set:

- 1,878 test samples,
- about 4.17 hours of audio.

Evaluation used:

- identical test split,
- identical text normalization,
- identical 16 kHz audio paths,
- identical evaluation code path,
- WER and CER as the main metrics.

The same decoding setup was used for the evaluated Whisper checkpoints, with:

- `language="arabic"`,
- `task="transcribe"`,
- `max_new_tokens = 225`.

The evaluation was then stratified by:

- duration bucket,
- speaker bucket,
- overlap bucket,
- turn-rate bucket,
- speech-rate bucket,

- text-length bucket,
- source dataset,
- source split,
- multiple-speaker and overlap flags.

This was important because the real question is not only *what is the global WER?*, but also *where does the model fail?*

12 Global Results

Table 4: Global comparison on the same curated test set

Model	WER (%)	CER (%)	Exact match (%)
Local Whisper Base	53.15	25.75	1.70
Local Whisper Small	40.52	19.37	6.18
`ychafiqui/whisper-small-darija`	49.49	23.41	0.27

12.1 Main Global Conclusion

The locally fine-tuned Whisper Small checkpoint is the best model in this study.

Compared with the local Whisper Base checkpoint, local Whisper Small improves by:

- 12.63 WER points,
- 6.38 CER points.

Compared with the public Hugging Face Darija small checkpoint, local Whisper Small improves by:

- 8.97 WER points,
- 4.04 CER points.

This is a strong result: under the same curated test protocol, the local data curation and fine-tuning pipeline clearly helped.

13 Bucketed Results and Interpretation

13.1 Duration Buckets

Table 5: Duration bucket comparison

Bucket	Samples	Base WER	Small WER	Interpretation
short_<3s	709	55.36	42.22	Small clearly better
medium_3_8s	816	59.76	44.92	Small clearly better
very_long_>=15s	347	50.80	38.89	Small clearly better
long_8_15s	6	117.57	87.84	Too small to trust

Small is better than Base in every duration bucket. The gain is not restricted to short clips; it also appears for very long clips.

However, the duration distribution is bimodal, so the "very long" bucket mostly reflects the long clipped source rather than a smooth long-form speech regime. The 8–15 second bucket contains only six samples and should not be over-interpreted.

13.2 Speaker and Overlap Buckets

Table 6: Speaker bucket comparison

Bucket	Samples	Base WER	Small WER	Small exact match
single_speaker	1692	52.88	40.63	6.68
two_speakers	186	53.86	40.24	1.61

Table 7: Overlap bucket comparison

Bucket	Samples	Base WER	Small WER	Small exact match
no_overlap	1694	52.88	40.61	6.67
light_overlap	149	53.65	39.02	2.01
heavy_overlap	35	54.75	45.36	0.00

These buckets are particularly interesting because they speak directly to the two original hypotheses.

The evaluation suggests that the usability filtering worked:

- the gap between single-speaker and two-speaker subsets is not dramatic after filtering,
- light-overlap clips remain manageable,
- heavy-overlap clips are still harder, but the sample count is very small.

This provides evidence that:

- mild backchannel behavior does not automatically make a clip unusable,
- multi-speaker clips with a dominant main speaker can still function as good ASR examples.

At the same time, overlap is not the main driver of the final test error because:

- 90.2% of the test set is `no_overlap`,
- only 35 samples fall into `heavy_overlap`.

13.3 Turn-Rate Buckets

Table 8: Turn-rate bucket comparison

Bucket	Samples	Base WER	Small WER	Comment
high_turn_rate	1735	53.85	40.91	Main regime
medium_turn_rate	140	50.26	38.31	Small better
low_turn_rate	3	173.68	167.37	Too small to trust

Again, Small is consistently better. The low-turn-rate bucket is too small for meaningful interpretation.

13.4 Speech-Rate Buckets

Table 9: Speech-rate bucket comparison

Bucket	Samples	Base WER	Small WER	Small exact match
medium_speech	1341	50.42	37.92	5.59
slow_speech	291	98.64	78.78	8.93
fast_speech	246	54.71	44.26	6.10

This is one of the clearest findings in the entire evaluation.

Slow speech is the hardest major regime for both models.

This is counterintuitive at first, since fast speech is often assumed to be harder. But in this dataset, the slow-speech bucket likely captures a mix of:

- long clips with sparse speech,
- long pauses,
- segmentation mismatch,
- weak alignment between audio and transcript,
- insertion-heavy decoding behavior.

So in this project, `slow_speech` seems to act more like a proxy for *long-context and alignment problems* than for easy slow articulation.

13.5 Text-Length Buckets

Table 10: Text-length bucket comparison

Bucket	Samples	Base WER	Small WER	Small exact match
short_text	1392	58.75	44.42	7.04
long_text	341	49.71	37.93	0.00
very_short_text	125	139.57	115.74	14.40
medium_text	20	61.19	49.75	Too small

Very short transcripts are the hardest text regime, and both models show WER well above 100 there. This usually indicates heavy insertion errors and/or transcript-audio mismatch.

This is one of the strongest signs that part of the remaining error is not just "the ASR model hears badly," but also:

- segmentation mismatch,
- transcript/audio mismatch,
- orthographic inconsistency,
- or out-of-domain content.

13.6 Source Dataset Buckets

Table 11: Source dataset comparison

Source	Samples	Base WER	Small WER	Small exact match
darija_tts_clean_metadata_full	1532	60.10	46.04	7.57
darija_speech_to_text_metadata_full	346	49.97	37.99	0.00

The speech-to-text source is easier than the TTS-clean source in this evaluation, despite being longer on average. This suggests that difficulty is not explained by duration alone. One likely reason is that the TTS source contributes many short clips where strict lexical matching and orthographic variation are heavily penalized.

13.7 Other Buckets

Some buckets are less informative because the final filtering already made the test set very clean:

- `quality_bucket` is entirely `quality_very_high`,
- `token_count_outlier_flag`, `remove_short_audio_long_text_flag`, `remove_token_outlier_flag`, `processing_error_flag`, and `diarization_empty_output_flag` all collapse to the clean case on the final test set,
- `multiple_speakers_flag` and `overlap_speech_flag` largely repeat the speaker and overlap findings above.

This is expected after a strict threshold of 0.85.

13.8 Original Source Splits

The `source_split_bucket` refers to the original split from the source dataset, not the final merged train/-validation/test split used for model training and evaluation. These buckets are useful for provenance, but they should not be over-interpreted as final experimental splits.

14 Comparison with a Public Hugging Face Checkpoint

I also evaluated the public model:

- `ychafiqui/whisper-small-darija`

On the same curated test set, it achieved:

- WER: 49.49
- CER: 23.41
- exact match: 0.27%

This is clearly below the locally fine-tuned Whisper Small checkpoint:

- local small WER: 40.52
- local small CER: 19.37
- local small exact match: 6.18%

This does not mean the local small checkpoint is universally better on every Darija ASR task. It means that under the same preprocessing, filtering, normalization, and evaluation protocol, the local training pipeline produced a stronger model for this curated benchmark.

15 Qualitative Error Analysis from Listening

Listening to the worst cases showed that not all large errors are "pure ASR failures." Several recurring patterns appeared:

1. **Heavy French inside Moroccan sentences.** These examples increase lexical variation and interact badly with non-standard transcript spelling.
2. **Very long audio, especially above 30 seconds.** These cases often led to missed content and much larger WER. Long chunks are especially vulnerable to segmentation mismatch and decoding incompleteness.
3. **Semantically reasonable prediction, but orthography not standardized.** In several cases, the prediction was close from a human perspective, but the written form differed from the reference. This is one of the biggest problems in Darija evaluation.

4. **Heavy music or strong non-speech content.** Some clips with music triggered outputs related to music rather than the intended transcript.
5. **English speech with Moroccan transcription labels.** Some clips contained full English audio paired with Moroccan-style transcripts. These are not normal ASR failures; they are dataset contamination or language mismatch cases.

15.1 What the Listening Study Suggests

A meaningful fraction of the worst errors come from:

- code-switching,
- annotation mismatch,
- orthographic instability,
- long-chunk segmentation problems,
- non-speech contamination,
- wrong-language content.

So the remaining error is not explained by model weakness alone.

16 What These Results Mean

16.1 Main Technical Result

Under the same compute budget and the same curated dataset:

- Whisper Small clearly outperformed Whisper Base,
- Whisper Small also outperformed the evaluated public Darija Whisper Small checkpoint.

16.2 Main Data Result

The most important bottleneck is *data quality and data definition*, not only model size.

The strongest evidence for this is:

- very high WER in `very_short_text`,
- very high WER in `slow_speech`,
- wrong-language and music-contaminated cases,
- orthographic near-misses that still count as full WER errors.

16.3 Main Linguistic Result

Moroccan Darija ASR remains difficult because the target language itself is unstable in written form.

The system is not only learning speech recognition; it is also forced to deal with:

- multiple spellings,
- mixed-language utterances,
- inconsistent transcript conventions,
- limited linguistic standardization.

So progress in Darija ASR is not only a modeling problem. It is also a data curation and language standardization problem.

17 Limitations

This project has several important limitations:

1. **Strict usability filtering.** Using a threshold of 0.85 greatly improved trainability, but also removed about 20.6% of samples and about 29.5% of total audio hours.
2. **Bimodal duration distribution.** The dataset contains many clips around 3–4 seconds and many around 25–30 seconds, with little in between.
3. **Small unreliable buckets.** Some strata are too small for strong conclusions, especially the 8–15 second bucket and the low-turn-rate bucket.
4. **Orthographic instability.** WER and CER are strict metrics and can over-penalize spelling differences that do not reflect true semantic failure.
5. **Code-switching and contamination.** Some clips contain heavy French, English, music, or transcript/audio mismatch.
6. **Single-machine budget.** With only 8 GB VRAM, the experiment could not explore larger checkpoints or a broader hyperparameter search.

18 If I Had More Time

If I had more time, I would do the following.

18.1 Data Improvements

- manually inspect and remove wrong-language clips,
- detect and remove music-heavy segments,
- add automatic language identification,
- improve segmentation for long 25–30 second chunks,
- re-check transcript/audio alignment.

18.2 Evaluation Improvements

- add a more relaxed normalization pipeline for Darija spelling variants,
- compute both strict and normalized WER/CER,
- build a manually validated clean in-domain Darija test subset,
- separate:
  - pure Darija,
  - Darija + French,
  - English contamination,
  - music/noise contamination.

18.3 Modeling Improvements

- compare against a CTC baseline such as Wav2Vec2,
- explore decoding controls for long-form stability,
- test whether slightly lower usability thresholds improve generalization,
- study whether some backchannel-heavy clips should be kept instead of filtered out.

18.4 Community / Linguistic Side

More broadly, Moroccan Darija would benefit from more community work on:

- orthographic conventions,
- transcript guidelines,
- linguistic normalization,
- code-switch annotation,
- standardized evaluation protocols.

Without that, ASR metrics will continue to mix true acoustic errors with writing-convention disagreements.

19 Conclusion

This project showed that a meaningful Darija ASR system can be built under tight compute constraints, but success depends more on *careful data curation* than on brute-force model scale.

The main contributions of the work are:

- selection of two complementary Moroccan Darija datasets,
- enrichment with diarization-derived metadata,
- design of a custom usability score based on overlap, dominant-speaker ratio, and turn-taking density,
- strict usability filtering at 0.85,
- training and comparison of Whisper Base and Whisper Small,
- evaluation against a public Darija Whisper checkpoint,
- detailed stratified error analysis.

The strongest result is clear:

> Whisper Small is the best model in this study, with 40.52 WER and 19.37 CER on the curated test set, outperforming both the local Whisper Base checkpoint and the evaluated public Hugging Face Whisper Small Darija checkpoint.

The broader conclusion is even more important:

> For Moroccan Darija ASR, the main bottleneck is not only model capacity. It is the combination of overlap speech, segmentation quality, code-switching, label inconsistency, and the lack of a standardized written form.

That is why the curation pipeline mattered so much in this project, and why future progress on Darija ASR will likely require improvements in both machine learning and language-resource standardization.